UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2011 question paper

for the guidance of teachers

5054 PHYSICS

5054/41 Paper 4 (Alternative to Practical), maximum raw mark 30

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1 (a) (i) capacitor is (fully) charged / can hold no more charge — B1 [1]

(ii) Z and no resistor / capacitor short circuited / current largest — B1 [1]

(b) 88 mA cao — B1 [1]

(c) (i) axes: labels correct way round, labelled quantity and unit — B1
scales: more than ½ grid, sensible — B1
y-axis: 2 cm ≡ 10 mA *x*-axis: 2 cm ≡ 10 s
points plotted accurately — B1
best fit smooth curve neatly drawn — B1 [4]

(ii) as *t* increases I decreases (non-linearly) / inversely related / exponential decrease — B1 [1]

(iii) 13 (mA) seen ± 1.0 — C1
1.3 V ± 0.10 ecf graph — A1 [2]

[Total: 10]

2 (a) (i) immerse stopper in water/can — B1
measure volume/collect water from spout — B1
measuring cylinder / balance to find mass hence volume — B1 [3]

(ii) diameter too small for stopper/object — B1
object not (fully) immersed — B1 [2]

(iii) any TWO sensible comments, e.g.:
wait for can to stop dripping before immersing stopper / filled exactly to spout
place stopper in without splashing / tie on thread / lower slowly
use sensitive measuring cylinder
stopper dry before immersing
measuring cylinder dry before use
use level bench
avoid parallax reading measuring cylinder
repeat and average — B2 [2]

(b) (i) mass — B1 [1]

(ii) balance / top-pan balance / beam balance — B1 [1]

[Total: 9]

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3 **(a)** **(i)** suitable arrangement of apparatus described or on diagram B1
reads meter / notes weight when just moves B1 [2]

(ii) sensible comment, e.g.:
increase force slowly / adds weights gently
use same part of bench
choice of newton meter described
repeat readings to find average
string horizontal
check for zero error in meter
sensible comment on friction over pulley B1 [1]

(b) use different sides of the same block B1 [1]

(c) **(i)** *F* / N and *W* / N B1 [1]

(ii) plot *F* against *W* (or *W* against *F*) / finds gradient of graph B1
gradient = *k* (or gradient = 1/*k*) B1 [2]

[Total: 7]

4 **(a)** **(i)** straight line from lamp to bench just above/touching top of card B1 [1]

(ii) correct indication of region of shadow B1 [1]

(b) shadow becomes longer B1 [1]

(c) multiple sources e.g.:
moon out
reflections
more street lamps
lights from other sources such as cars/houses
larger lamp size B1 [1]

[Total: 4]

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